

Network

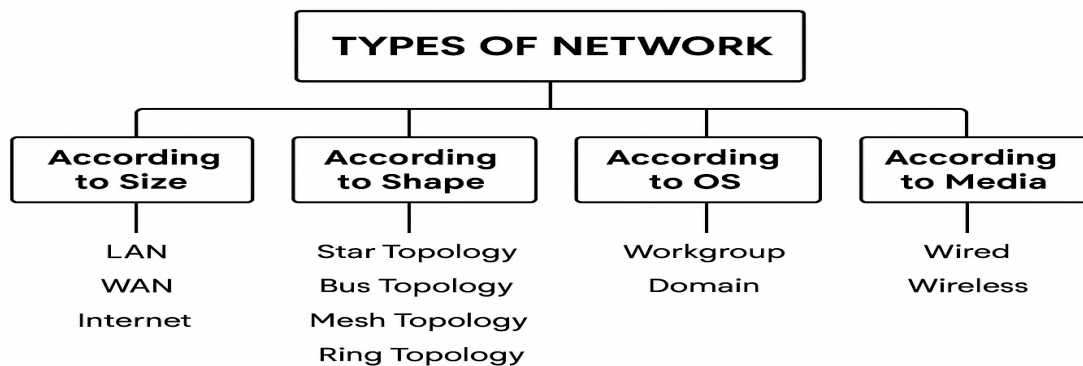
Network

What is the network?

A **network** is a group of two or more computers or devices connected together so they can share information, files, and resources.

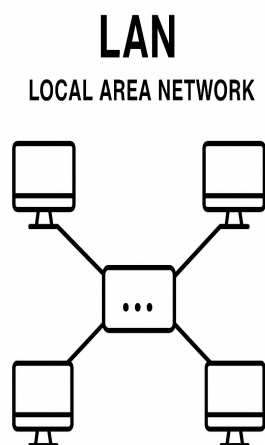
What is networking ?

Networking is the process of connecting two or more computers or devices together so they can communicate, share data, and use resources.



LAN (stand for local area network)

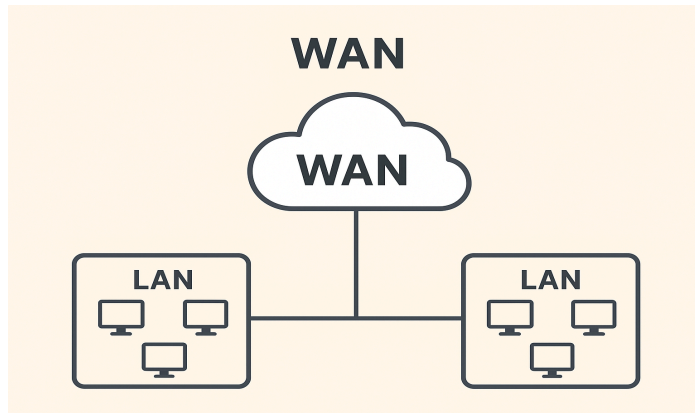
It is a network used to communicate between computers inside the same building or area



WAN (Wide Area Network) When two or more LANs are connected through the Internet, we call it a WAN
Or

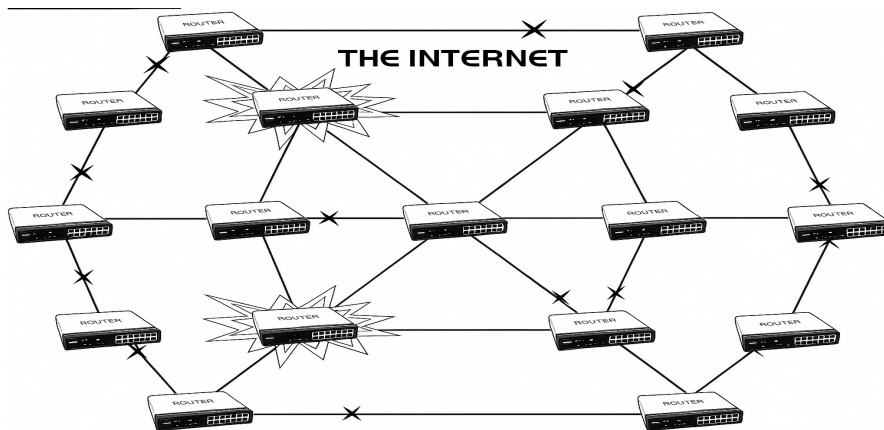
It is a network made by connecting many LAN networks together across different cities, states, or countries.

Example: internet is a wide area network



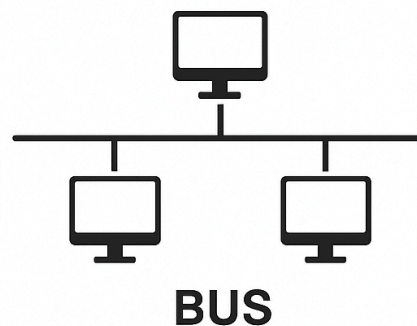
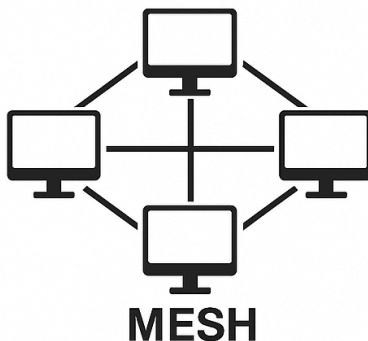
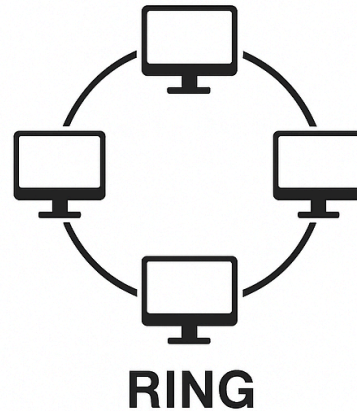
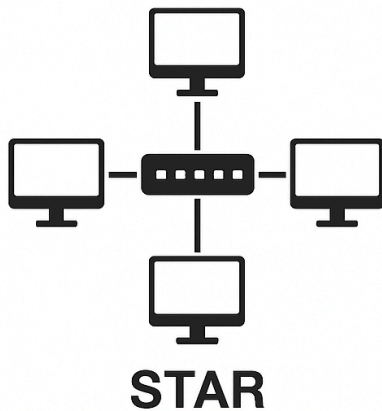
What is the **internet**?

It is a group of many LAN and WAN networks that are connected to each other and share resources.



ACCORDING TO SHAPE

According to shape we have four types of network.



1. Star topology:

Every devices connect to the central switch or hub

Advantages : 1) All computers can communicate with each other through the central Switch or Hub.

2) If one computer gets disconnected, it does not affect the communication of other devices.

Drawbacks: 1) Whenever computers communicate with each other, the message is always broadcast by the hub or switch.

2) If the central switch or hub fails, then the entire network will stop working.

2. Ring topology

All computers are connected in a circle via cable, and they can all communicate with each other

Advantages : 1) all computers can communicate with each other.

Drawbacks : 1) If the cable breaks, the whole network stops working.

2) It is slower than bus topology because the data has to pass through many computers, which creates traffic.

3. Bus topology

All computers are connected using one straight main cable, and they can all communicate with each other.

Advantages : 1)All computers can communicate with all other computers
2)cheap and best to create a small network

Drawbacks:1)if the main cable breaks, the whole network stops.

4. Mesh topology:

All computers are connected to every other computer directly, so every computer can communicate with every computer.

Advantages :

- 1)It is highly reliable because if one connection fails, communication can continue through the other existing paths.
- 2)Every computer is directly connected to all other computers, so data can travel quickly without waiting for intermediate devices.

Drawbacks:1)It is high in cost

Because we have to attach many cables in this network, the setup becomes very expensive.

2)It is not easy to manage when devices increase

If the number of devices increases, the number of connections also increases, which makes the network difficult to manage.

We have two types of addresses in networking

1. Physical Address (MAC Address)

A Physical Address is the hardware address of a network device, permanently assigned to its network interface card (NIC) by the manufacturer.

- **MAC Address → Media Access Control Address**
- It is a 12-digit hexadecimal number
- Used inside a local network (LAN) to identify devices
- Helps data reach the correct device within the same network.
- It is burned into the NIC at the time of manufacture.

2. Logical Address (IP Address)

What is IP Address

'IP' stands for 'Internet Protocol'. Each device that connects to the Internet needs a unique identifying number with which to communicate, called an 'IP address'.

- It is Used to identify devices globally, beyond the local network.
- It will help us to find the correct path across multiple networks.
- It can be static (manually set) or dynamic (automatically assigned by DHCP).
- You can change your IP address easily.

Versions of IP Address

There are two versions of IP

-IPv4

Ex 192.168.1.5

•IPv6

Ex 2001:0db8:85a3:0000:0000:8a2e:0370:7334

IPv4 Addressing

IPv4 (Internet Protocol Version 4):

- Supports about 4.3 billion unique addresses.
- Uses a 32-bit address
- Consists of four Octets (1 Octet= 8Bits)
- Each octet separated by dots (.)
- Represents in Decimal notation

Example of IPV4 Addressing

Let's see an example of IPv4 address

172 . 20 . 200 . 1
10101100.00010100.11001000.00000001

Total class of IPv4Addressing

1. Class A
2. Class B
3. Class C
4. Class E
5. Class D

Uses of all ipv4 classes

1. CLASS A: It is generally used by large organizations ISPs.
2. CLASS B: It is typically used by mid-size companies and academic institutions.

- 3.CLASS C: This class is commonly used by small businesses or for home networks
- 4.CLASS D: Class D is reserved for multicasting
- 5.CLASS E: Class E addresses are reserved for experimental and future

Range of IPV4 Address Classes

Address Class	RANGE	Default Subnet Mask
A	1.0.0.0 to 126.255.255.255	255.0.0.0
B	128.0.0.0 to 191.255.255.255	255.255.0.0
C	192.0.0.0 to 223.255.255.255	255.255.255.0
D	224.0.0.0 to 239.255.255.255	Reserved for Multicasting
E	240.0.0.0 to 254.255.255.255	Experimental

Note: Class A addresses 127.0.0.0 to 127.255.255.255 cannot be used and is reserved for loopback testing.

IPV4 total ip addresses:

$256 \times 256 \times 256 \times 256 = 429,49,67,296$

IPV4 (START AND END POINT)

0.0.0.0 TO 255.255.255.255

Class A:(start & End point)

0.0.0.0 to 127.255.255.255

Note : 1)The ip address range 127.0.0.0 to 127.255.255.255 is reserved for loopback and not used for public and private networking ,even though it technically falls under Class A..

127.0.0.0- loopback addresses

Purpose : *Used by a device to send network traffic to itself for testing and diagnostics.*

Most common: *127.0.0.1 is often called "local host".*

2)"0.0.0.0 to 0.255.255.255 we cannot use for host computers because this range is reserved in class A.

It is used only when a device has no IP or for the default route.

So we never give these IPs to any host."

Class A private ip address range:

Start from

10.0.0.0 to 10.255.255.255

What is subnet mask

It will help us to find the how many network bits reserved for network inside our ip address

OSI MODEL

1) OSI model stand for Open System Interconnection

2) OSI model was created by **ISO(international standard organization)** in 1984 to make different computers communicate using a common standard

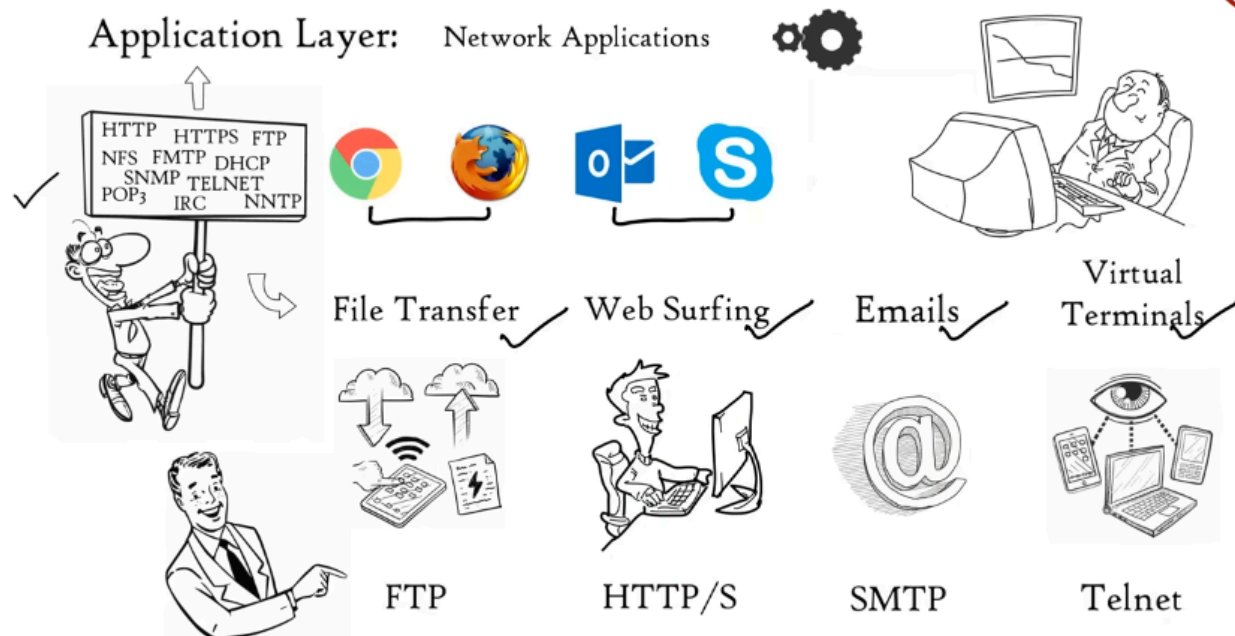
3) ISO created the OSI Model so all computers in the world can communicate using one common standard."

4) OSI Model explains **how data moves** from one device to another device in 7 steps (layers). Each layer has its own job.

5) we also call it a reference model of **tcp/ip.(introduced in 1974)**

We have 7 layer of osi model as i have mention below:

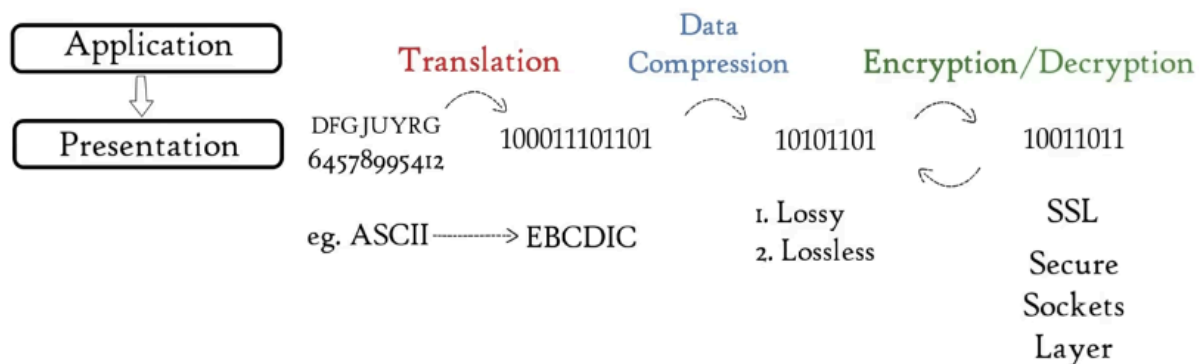
Application layer



- Application layer is used by network application
- Network application means like google chrome,firefox,or we can consider a application which are run on your system with the help of internet
- If i want to transfer file we used **FTP(FILE TRANSFER PROTOCOL)**
- For web surfing we use**HTTPS(hyper text transfer protocol)(443)**
- For sending email we used **SMTP(simple mail transfer protocol)(465)**
- For accessing virtual terminal we use **TELNET**
- **It's depends on user activity**

Presentation layer

Presentation Layer



- Presentation layer received data from application layer in the form of character and numbers

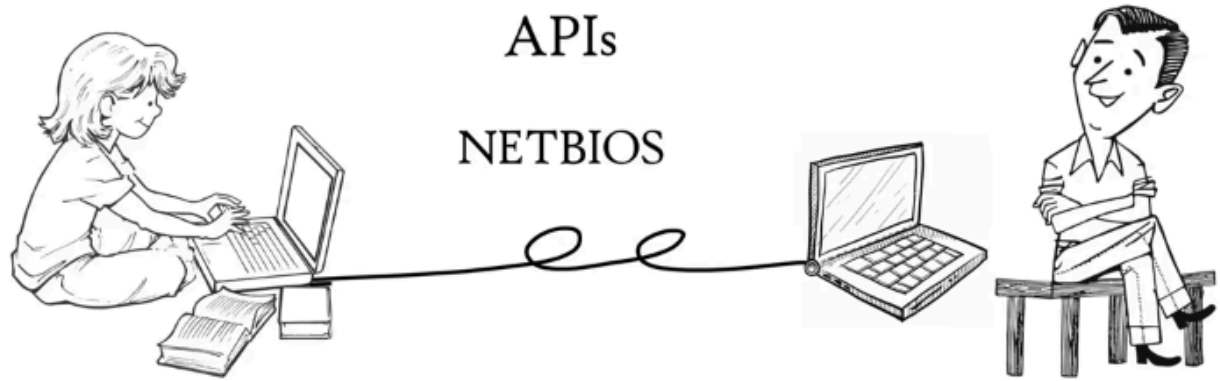
- Presentation layer converts these data into the machine understandable language(binary 0's & 1's)
- Before data is transmitted, the presentation layer reduces the number of bits that i used to represent the original data this bit reduction process is known as data compression. And it can be lossy or lossless.
- Data compression is used to reduce the amount of space used to store the original file as the size of file reduce it can be received at destination very less time that is data transmission can be done faster
- To maintain integrity of data before transmission data is encrypted with the help of **SSL(secure socket protocol)**
- Encryption enhances the security of the sensitive data .At sender side data encrypted and at receiver side data will be decrypted.
- So presentation layer perform three basic operation translation ,compression ,encryption&decryption

Session layer

- Session layer will help us to set up and manage connection enabling sending and receiving of data followed by transmission of connection or we can say session.
- For example, suppose you have a plan for a party and you have hired a few helpers to ensure that each activity runs smoothly.
- The helper will help you in setting up, cleaning up and closing the party .
- Same like in your party you had some helper inside your session layer has two own helpers called (API's) Application Programming Interface and netbios.

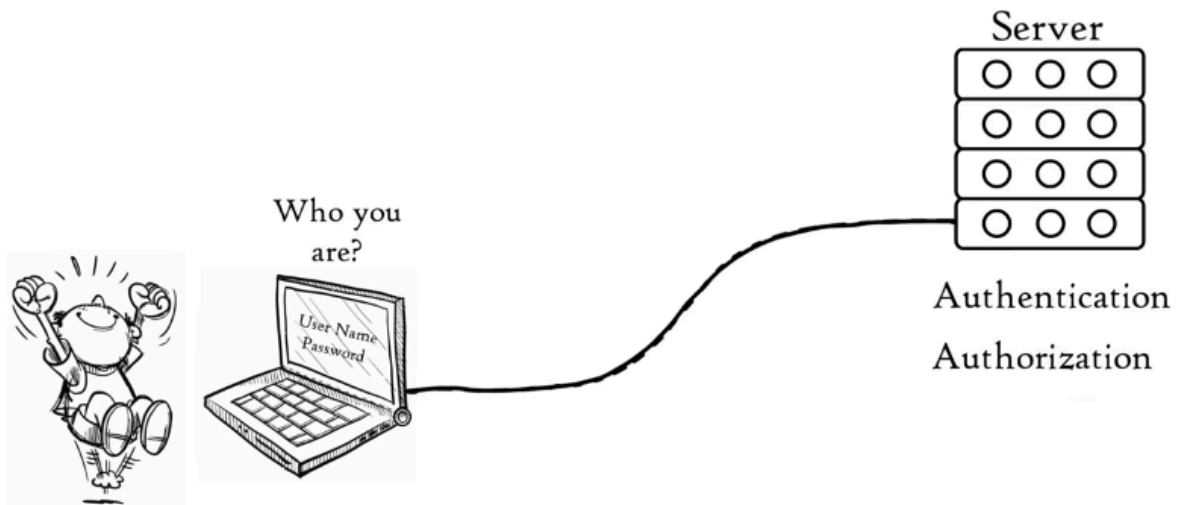
- Netbios is an example of an api which allows applications on different computers to communicate with each other.

Session Layer



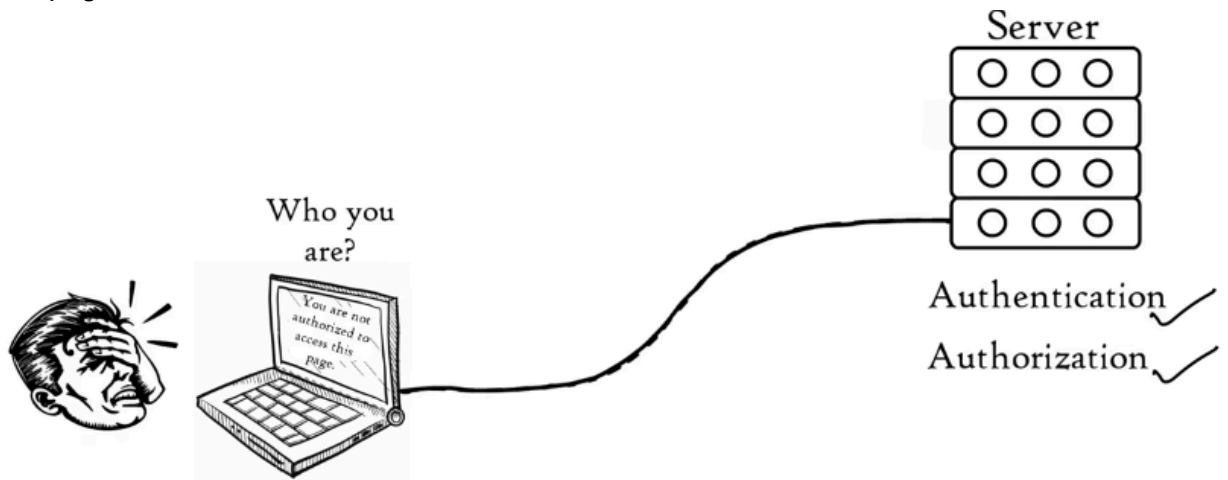
- Session management
- Authentication
- Authorization

- Just before a session or connection is established with the server. The server performs a function called authentication and authorization .

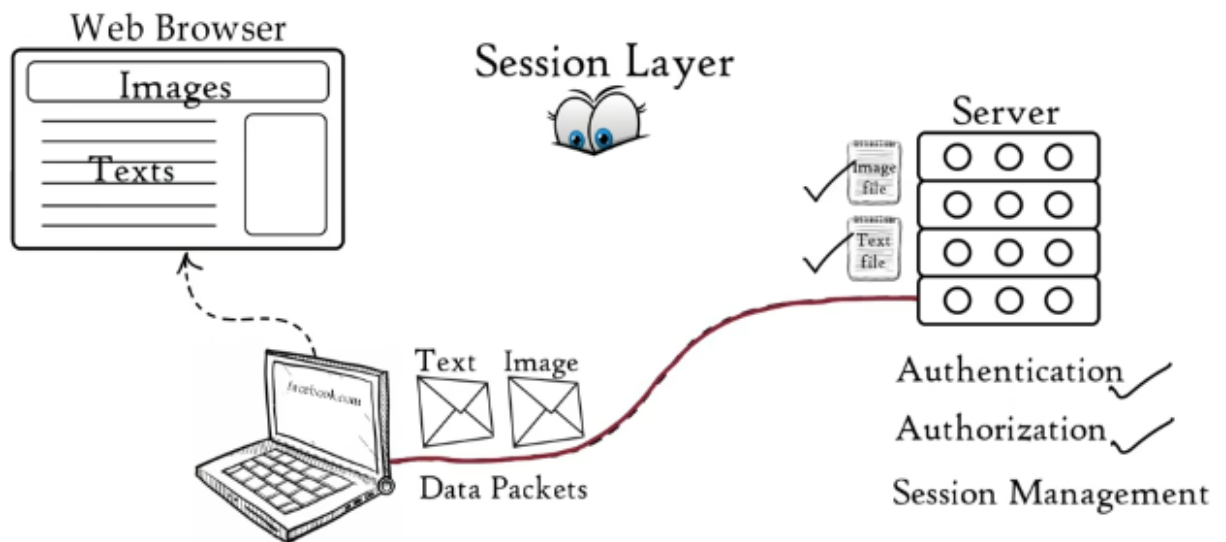


- **Authentication** is the process to verifying who are you ?(user is valid or not)
- To perform authentication, the server displays the user and password cursor box on the host computer.
- Once a user enters password and user name & if the password is matched then the session or connection will be established.

- After established connection authorization will be checked.
- **Authorization is** the process used by the server to determine if you have permission to access this file; if not you will get a message saying you are not authorized to access this page.



- Both of these function **authentication** and **authorization** are performed by the session layer.
- A **new session** is created between your device and the server



- This session keeps the connection active
- After the download finishes, the session is **closed**.

Transport Layer

- Transport layer control the reliability of communication

- **Segmentation:** The data received from the session layer is divided into smaller data units called segments.
- **Flow control:** transport layer control the amount of data being transmitted
Ex: consider mobile connected to a server and suppose server can transmitted data max 100 mbps
And our mobile can process data maximum 10 mbps
Now we are downloading a file from the server and server start sending data at 50 mbps which is grater that the rate of our mobile can process so mobile phone with the help of transport layer tell the server slow down data transmission rate up to 10 mbps so that no data get lost
Similarly, the server sending data at 5 mbps rate then mobile phone tells the server to increase data transmission rate up to 10 mbps to maintain system performance
- **Error control:** transport layer also helps in error control if some data segments do not arrive at their destination .
- The transport layer uses an automatic repeat request process to retransmit the loss data packet.

Protocols of transport layer

1)TCP (transmission control protocol):connection oriented transmission.

2)UDP(user datagram protocol):connection less transmission.

• **Udp is faster than TCP**

BECAUSE:UDP does not provide any acknowledgement or feedback.

Transport Layer is involved in

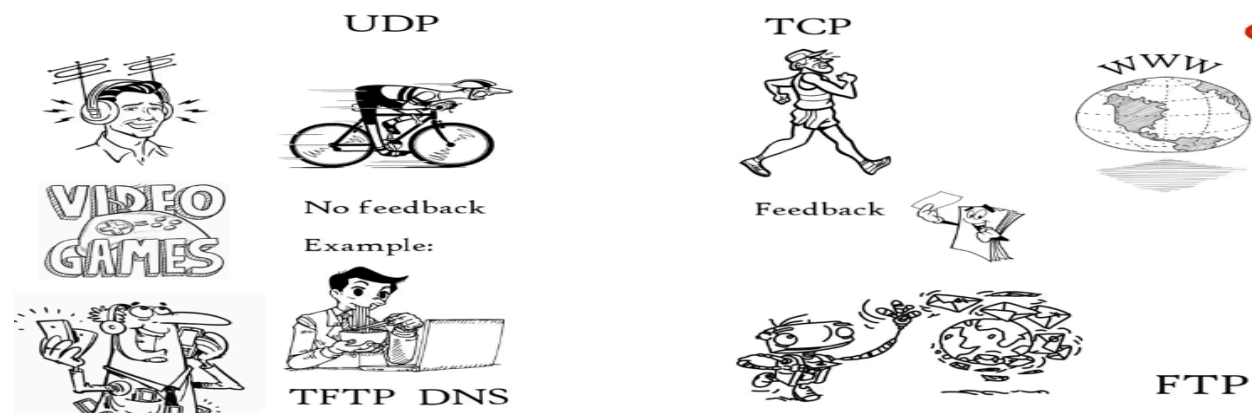
Segmentation

Flow control

Error control

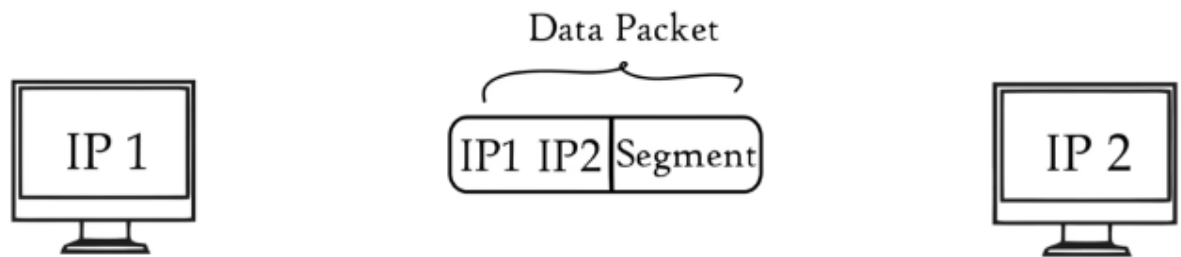
Connection oriented transmission.

Connection less transmission.

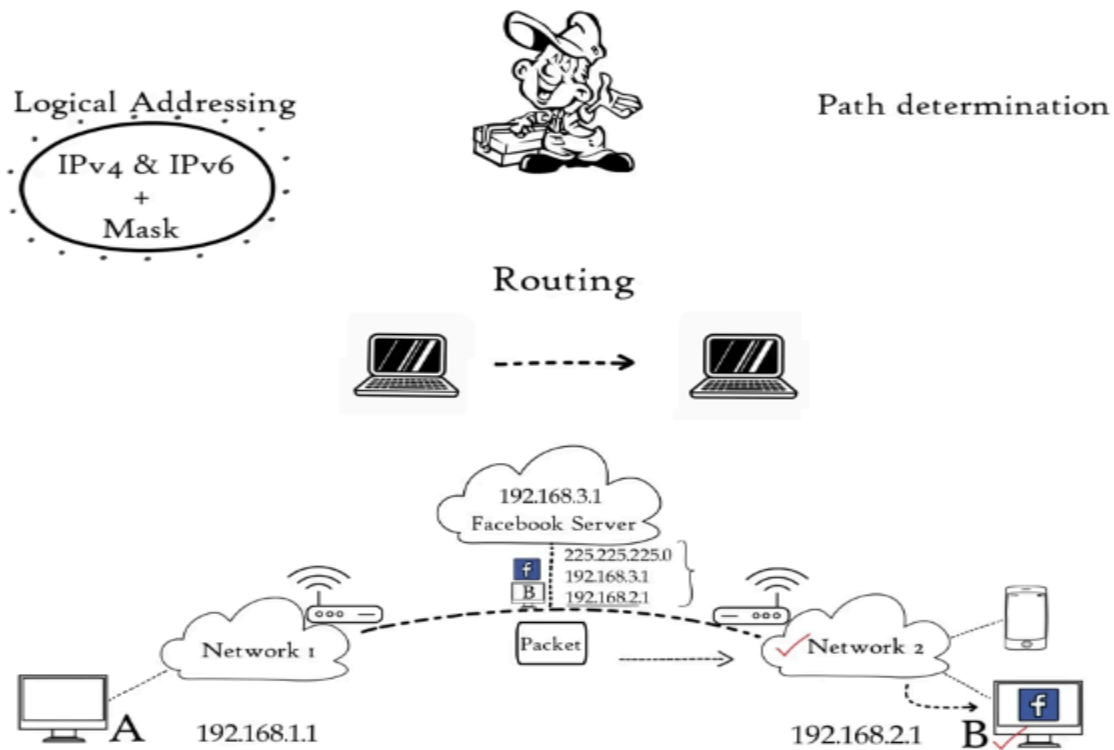


- The network layer is responsible for transmitting the data packets that it receives from the transport layer.

- It forwards these packets from the source computer to the destination computer, even if they are located in different networks.
- The data unit at the network layer is called a **packet**.
- IP addressing done in the network layer is known as **logical addressing**.
- Every device in a network has a unique IP address.
- The network layer adds the sender's and receiver's IP addresses to each data packet.



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- **Routing** is the process of moving data packets from the source to the destination, and it works based on IPv4 or IPv6 addresses.

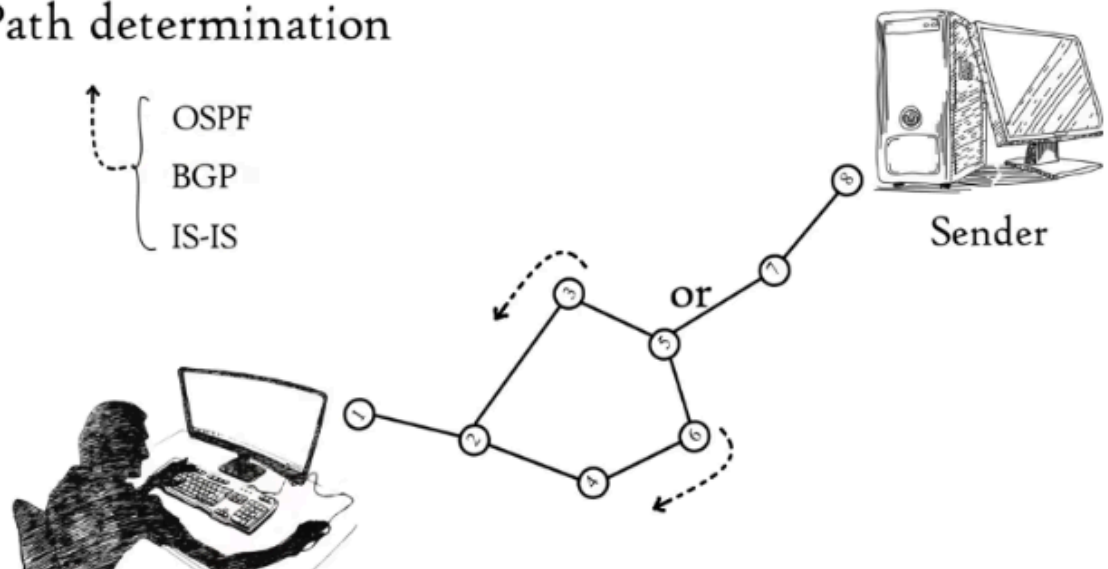


Path determination:

Path determination is the network layer's process of choosing the best route for data packets to travel across the network using routing protocols.

Path determination is the network layer's process of choosing the best route for data packets to travel across the network using routing protocols.

Path determination



Private ip ranges

Class a 10.0.0.0 to 10.255.255.255

Class b 172.16.0.0 to 172.31.255.255

Class c 192.168.0.0 to 192.255.255.255

Private ip it is provided by system administrator

Private IP is used to assign host computers which are connected inside the LAN network.

Public ip address

It is provided by isp(internet service provider)

It is used to access resources globally

Sender	receiver
Application	application
Presentation	presentation
Session	session
Transport	transport
Network	network
Data link	datalink
Physical	physical

tcpip/1974
Application
Transport
Network
Physical

- 1)What is the difference between tcpip modal and osi model**
- 2)osi model layers and it's work**
- 3)router , hub and switch these devices work in which layer**

Classes 5

Class A ----- 0 to 127 -----(10.0.0.0 to 10.255.255.255)1 to 9& 11 to126

Class B -----128 to 191-----172.16.0.0 to 172.31.255.255

Class C-----192 to 223-----192.168.0.0 to 192.168.0.255

Class D-----224 to 239

Class E-----240 to 255

125.0.0.0 network id

125.0.0.1

125.0.0.2

125.0.0.3

125.0.0.255 broadcast id

125.0.1.0

125.0.1

255.0.0.0 2^{24} =

193.255.123.0

255.255.255.0

Interview questions

Basics

1. What is networking, and what are its types (LAN, WAN, MAN)?

Networking is the process of connecting two or more computers or devices together so they can communicate, share data, and use resources.

2. What is an IP address? Difference between IPv4 and IPv6?

IPv4 and IPv6 are used to identify devices on a network. IPv4 uses a 32-bit address, for example, 192.168.1.10. IPv6 uses a 128-bit address, for example, 2001:db8::1. IPv6 was introduced because IPv4 addresses are limited. IPv6 provides many more IP addresses and supports the growing number of devices on the Internet

3. What is a MAC address, and how is it different from an IP address?

A MAC (Media Access Control) address is a physical address that is permanently assigned to a network interface card (NIC) by the manufacturer. It is used for communication within the same local network (LAN).

An IP (Internet Protocol) address is a logical address assigned to a device either manually or by a DHCP server. It is used to identify a device and communicate across different networks, including the Internet.

4. What is DNS?

DNS stands for Domain Name System. It is a service that translates a human-readable domain name, such as www.google.com, into an IP address, such as 142.250.183.68, so that computers can communicate with each other over the network.

5. What is DHCP, and why is it used?

DHCP stands for Dynamic Host Configuration Protocol. It is a service that automatically assigns IP addresses and other network settings like the subnet mask, default gateway, and DNS server to client computers. This eliminates the need to configure IP addresses manually. DHCP uses UDP port 67 on the server side and UDP port 68 on the client side.

Imagine your office has 100 computers.

Without DHCP: You must manually configure the IP address on all 100 computers.

With DHCP: When a computer connects to the network, it automatically receives an IP address and other network settings from the DHCP server.

DHCP follows the DORA process:

- **D – Discover:** The client broadcasts a request to find a DHCP server.
- **O – Offer:** The DHCP server offers an available IP address.
- **R – Request:** The client requests the offered IP address.
- **A – Acknowledge:** The DHCP server confirms the lease and assigns the IP address.

6. What is the difference between a router, switch, and hub?

OSI & TCP/IP Model

7. What is the OSI model? Name all 7 layers.

8. What is the TCP/IP model, and how does it differ from OSI?

The TCP/IP model is a practical networking model used for communication over the Internet. It has 4 layers. The OSI model is a conceptual reference model used to understand how network communication works. It has 7 layers.

9. What happens at each layer when data is sent over a network?

When data is sent over a network, it starts at the Application layer, where the data is created. The Transport layer adds port numbers and prepares the data. The Internet layer adds the source and destination IP addresses. The Network Access layer adds the MAC addresses and sends the data over the network. At the destination, each layer removes its own information, and the application receives the original data.

Protocols

10. What is the difference between TCP and UDP?

TCP is a reliable protocol. It makes a connection before sending data and makes sure the data reaches the destination. UDP is a fast protocol. It sends data without making a connection and does not check whether the data reaches the destination. We use TCP when data is important, and UDP when speed is important

<https://chatgpt.com/share/6a75bfdd-547c-83ee-b0fe-dcfb841dfa7e>

11. What is a three-way handshake in TCP?

The TCP Three-Way Handshake is the process used to create a connection between a client and a server before sending data. It makes sure both devices are ready to communicate

12. What is HTTP vs HTTPS?

HTTP is used to transfer web pages between a browser and a web server, but it does not encrypt the data. HTTPS is the secure version of HTTP. It encrypts the data using SSL/TLS, which protects sensitive information such as usernames, passwords, and banking details. HTTP uses port 80, while HTTPS uses port 443.

13. What is FTP, and what is it used for?

14. What is the difference between HTTP and HTTPS in terms of security?

HTTP stands for HyperText Transfer Protocol.

HTTPS stands for HyperText Transfer Protocol Secure.

The main difference is that HTTPS encrypts the data between the user's browser and the web server, while HTTP sends data in plain text.

Addressing & Ports

15. What is a subnet mask, and why is it used?

A subnet mask is a number that works with an IP address. It tells us which part of the IP address is the network and which part is the host (device).

We use a subnet mask to determine whether the destination device is on the same network or a different network. If it is on a different network, the data is sent to the router (default gateway).

16. What is NAT (Network Address Translation)?

NAT is a process used by a router or firewall to translate private IP addresses into a public IP address so that devices on a private network can access the Internet.

17. What are ports? Give examples of well-known ports (80, 443, 21, 22, 25).

Security

18. What is a firewall, and how does it work?

A firewall is a network security device or software that monitors and controls incoming and outgoing network traffic.

It allows trusted traffic and blocks unauthorized or harmful traffic based on predefined security rules.

Its main purpose is to protect computers and networks from unauthorized access and cyber attacks.

19. What is a VPN, and why is it used?

VPN stands for Virtual Private Network.

A VPN creates a secure and encrypted connection between your device and a remote network over the Internet.

We use a VPN to access office resources securely from outside the office and to protect data while it travels over the Internet.

Troubleshooting

20. What is the ping command used for?

The ping command is used to check network connectivity between two devices.

21. What is tracert, and how does it help diagnose network issues?

Tracert is a Windows command-line tool used to find the path that data packets take from your computer to a destination.

It displays each router (hop) between your computer and the destination, along with the time taken to reach each hop.

22. What is the difference between public ip and private ip?

A private IP address is used inside a local network and cannot be accessed directly from the Internet. It is assigned by a router or DHCP server and can be reused in different networks. A public IP address is assigned by an ISP, is globally unique, and is used for communication over the Internet. Devices with private IP addresses access the Internet through NAT

23. Why do we use private IP addresses?

Private IP addresses are used to identify devices within a local network while allowing them to share one public IP address for Internet access.

24. Can two devices have the same public IP?

No. A public IP address must be globally unique on the Internet.

25. Can two computers have the same private IP?

Yes, if they are in different networks. For example, your home network can have 192.168.1.10, and another company's network can also have 192.168.1.10.

26. Why is TCP slower than UDP?

Because TCP establishes a connection, sends acknowledgments, performs error checking, retransmits lost packets, and maintains packet order.

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<https://www.youtube.com/@PowerCertAnimatedVideos>

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